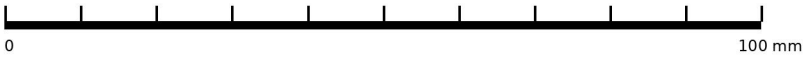
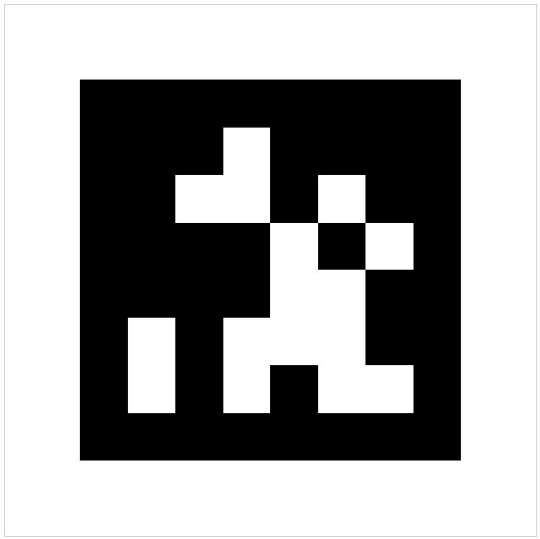
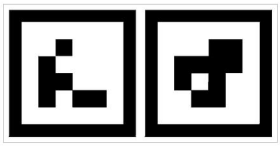
Wuji pen-spin markers - print at 100% scale (no 'fit to page'). Check: the ruler below must measure exactly 100 mm.





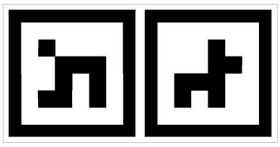
Wrist tag: AprilTag 36h11, ID 0 - black square edge = 50.4 mm (keep the white margin). Mount it as in Wuji's assembly photo.

Pen end-block faces (sticker fallback). Each block is 18 x 18 x 36 mm; the 36 mm side runs along the pen. Faces are stretched to size on purpose.



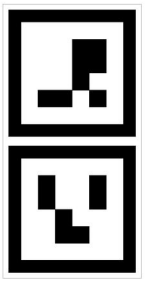
A_bottom +X

36x18 mm, ids 0 1



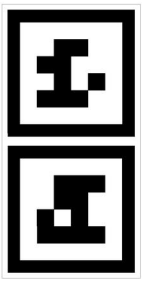
A_bottom -X

36x18 mm, ids 2 3



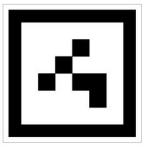
A_bottom +Y

18x36 mm, ids 4 5



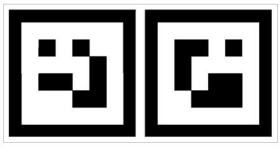
A_bottom -Y

18x36 mm, ids 6 7



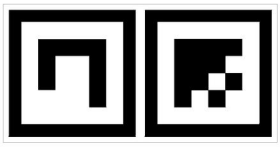
A_bottom -Z

18x18 mm, ids 8



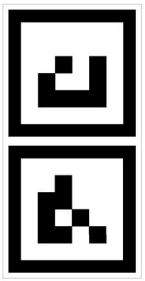
A_top +X

36x18 mm, ids 9 10



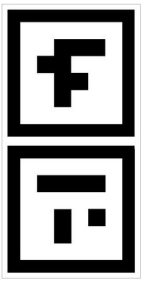
A_top -X

36x18 mm, ids 11 12



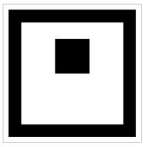
A_top +Y

18x36 mm, ids 13 14



A_top -Y

18x36 mm, ids 15 16



A_top +Z

18x18 mm, ids 17

Assembly (Wuji's setup guide): pen total length 250 mm; shaft core diameter 14.9 mm; 178 mm of shaft exposed between the two blocks; block centres 107 mm from the pen midpoint.

A_top block: its +Z face (ids 17) is the pen's tip end. A_bottom block: its -Z face (ids 8) is the other end. Print the textures exactly as shown, matte paper, no gloss.

Best option: 3D-print pen_heads.3mf (two-colour) + a shaft from the asset bundle instead of stickers; the policy was trained on that pen's size and mass.

Camera: any calibrated camera works through the observer's --source video; a RealSense colour stream can use its factory intrinsics (deploy/README.md).